



Sealed Pick

A party game for three on Midnight. Everyone seals a pick through a Compact contract; the seals break together.

THE PROBLEM

Hidden, simultaneous picks need a referee.

Coordination games, blind votes and sealed guesses only work when two rules hold:

- Nobody sees another pick before choosing.
- Nobody changes a pick after seeing someone else's.

Online, a server usually holds every pick, and everyone has to trust it.

Three players, three seals, one reveal.

01

Seal

Each phone keeps its choice and a random nonce in private state. The commit circuit proves the player owns the seat and the choice is in range, and discloses only the seal.

02

Wait

The public ledger shows three seals and a counter. No choice exists on chain yet, so there is nothing to peek at.

03

Break

After the last seal, the openings go to one player. The reveal circuit checks all three against their seals and publishes the choices and the score in one transaction.

What the ledger sees.

On each phone, private state

- The player's random 32-byte secret, which proves seat ownership
- The choice and its nonce until the reveal

On the public ledger

- Room ID, option count and phase
- Seat pseudonyms and seals
- After the reveal: the three choices and the score

```
pseudonym = persistentHash(domain, roomId, secret)
seal = persistentCommit((domain, roomId, seat, choice), nonce)
```

BUILT AND CHECKED

Running, tested, proven on a devnet.

Compact 0.31.1

Four circuits compile with proving and verifying keys.

33 tests pass

Rules, every commit order, all 64 answer triples, and a raw-ledger check that no choice appears early.

Browser table

Runs the generated circuits in the page. A forged second commitment is refused by the circuit.

```
npm run devnet:e2e · local devnet · real proofs
```

deploy	block 9	2887e469...969a	SucceedEntirely
createRoom	block 13	a58879f2...19ff	SucceedEntirely
joinRoom	block 17	ce2ffabc...ccd6	SucceedEntirely
joinRoom	block 20	8787d251...fa54	SucceedEntirely
commit	block 24	045b2e4f...9a77	SucceedEntirely
commit	block 27	ac972efa...6baf	SucceedEntirely
commit	block 31	181c27fd...bad5	SucceedEntirely
reveal	block 35	b10a4671...73c9	SucceedEntirely

NEXT

Where Sealed Pick goes from here.

Who plays

Friends on a call, classrooms learning about focal points, stream audiences voting without seeing the running tally.

What we build next

Deploy to Midnight Preview once a funded wallet is available. Send openings between phones over an encrypted channel. Room links and more seats.

How it pays

Free to play. Paid question packs, and branded rooms for streamers and events.

TRY IT

Three phones, one ledger, no peeking.

```
npm ci && npm run build && npm run web:dev
```

Then open localhost:5391 and press “Play a round for me”, or play all three phones yourself.